

1. Nitrate danger board

WHAT YOU SEE

Red 'NITRATE DANGER BOARD' poster left

WHAT IT ENCODES

Nitrates relieve ischemic pain but are CONTRAINDICATED/harmful in specific MI situations because their preload-reducing venodilation can cause catastrophic hypotension. The four classic 'danger board' entries: SBP <90 (or ≥ 30 below baseline), RIGHT-ventricular/inferior infarct, recent PDE5 inhibitor, and severe bradycardia/tachycardia. Nitrates never reduce mortality — they are purely symptomatic. Mnemonic: 'nitrates drop the preload; some hearts can't afford it.'

BOARD TRAP

WRONG answer: stating nitrates are safe in every MI. Discriminator — in preload-dependent states (RV infarct, hypotension) or with recent PDE5 inhibitors, nitrates can precipitate profound shock; they are symptomatic only and never mandatory.

2. SBP <90 mmHg

WHAT YOU SEE

BP cuff '<90 mmHg' top panel

WHAT IT ENCODES

Hypotension (SBP <90 mmHg, or a fall ≥ 30 mmHg from baseline) is an absolute contraindication to nitrates — their venodilation further drops preload, cardiac output, and coronary perfusion pressure, worsening shock and ischemia. Check the BP before EVERY nitrate dose. Mnemonic: 'low pressure + nitrate = floor.'

BOARD TRAP

WRONG answer: giving nitroglycerin regardless of blood pressure to relieve chest pain. Discriminator — in a hypotensive MI patient, nitrates deepen the hypotension and can cause cardiovascular collapse; treat the low pressure (fluids/pressors as indicated) and withhold nitrates.

3. RV infarct figure

WHAT YOU SEE

On the red NITRATE DANGER BOARD, the body/heart panel (RV-inferior infarct) with a red X over a nitro IV bag — nitrates are crossed out because this preload-dependent heart can't afford the drop.

WHAT IT ENCODES

Right-ventricular infarction (usually from proximal RCA occlusion accompanying an INFERIOR STEMI — ST elevation in II/III/aVF, confirmed by ST elevation in right-sided lead V4R) is exquisitely PRELOAD-DEPENDENT. The stunned RV cannot fill the LV without high venous return, so nitrates (which slash preload) cause profound hypotension. Treatment is the opposite: IV FLUID loading (and avoid nitrates/diuretics/morphine). Mnemonic: 'RV infarct = fill it, don't drain it.'

BOARD TRAP

WRONG answer: giving nitrates first-line for chest pain in an inferior/RV infarct. Discriminator — the hypotensive inferior-MI patient may have RV involvement; nitrates cause severe hypotension here, whereas IV fluids are the correct first step. Always get a right-sided ECG (V4R) in inferior STEMI.

4. PDE5 inhibitor 24H

WHAT YOU SEE

On the danger board, a pill panel labelled '24H' with a red X over the nitro bag — short-acting PDE5 inhibitors (sildenafil/vardenafil) block nitrates for 24 hours.

WHAT IT ENCODES

Recent phosphodiesterase-5 inhibitor use is a hard contraindication to nitrates: both raise cGMP/NO-mediated vasodilation, and the combination causes severe, refractory hypotension. Withhold nitrates for ≥ 24 hours after SHORT-acting PDE5 inhibitors (sildenafil, vardenafil). Always ask the chest-pain patient about erectile-dysfunction/pulmonary-hypertension drugs. Mnemonic: 'sildenafil + nitrate = synergistic crash.'

BOARD TRAP

WRONG answer: claiming PDE5 inhibitors don't interact with nitrates. Discriminator — co-administration causes potentially fatal hypotension via shared cGMP pathway; you must take a medication history and respect the washout window before any nitrate.

5. PDE5 48H

WHAT YOU SEE

On the danger board, a pill panel labelled '48H' with a red X over the nitro bag — long-acting tadalafil extends the nitrate washout to 48 hours.

WHAT IT ENCODES

For the LONGER-acting PDE5 inhibitor TADALAFIL, the nitrate washout is ≥ 48 HOURS (vs 24 h for sildenafil/vardenafil) because of its longer half-life. Knowing which drug the patient took determines the safe interval. Mnemonic: 'Tadalafil = Two days (48 h); the others = one day (24 h).'

BOARD TRAP

WRONG answer: assuming a 24-hour washout is enough for all PDE5 drugs. Discriminator — tadalafil's long half-life requires 48 hours; giving nitrates at 24 hours after tadalafil can still produce dangerous hypotension.

6. Nitro is symptomatic only

WHAT YOU SEE

The referee on the conveyor line chewing a NITRO tablet — nitroglycerin works the line for symptom relief only, never crossing the mortality finish line.

WHAT IT ENCODES

Nitroglycerin relieves ischemic chest pain (venodilation lowers preload/wall stress; modest coronary dilation) but has NO mortality benefit after MI — proven by GISSI-3 and ISIS-4. It is an antianginal/symptomatic and hemodynamic agent only. The mortality-reducing STEMI adjuncts are aspirin, P2Y12 inhibitors, beta-blockers, ACE inhibitors, high-intensity statins, and (if EF<40 with HF/DM) aldosterone antagonists — plus reperfusion. Mnemonic: 'nitro for the pain, not the prognosis.'

BOARD TRAP

WRONG answer: stating nitrates lower mortality after MI. Discriminator — ISIS-4/GISSI-3 showed no mortality benefit; mortality-improving adjuncts are antiplatelets, beta-blockers, ACEi, statins, and aldosterone antagonists (and reperfusion), not nitrates.

7. Beta-blocker truth wall

WHAT YOU SEE

'BETA BLOCKER TRUTH WALL' green board

WHAT IT ENCODES

Beta-blockers reduce post-MI mortality, reinfarction, and arrhythmia by lowering HR, BP, and myocardial O₂ demand and raising the VF threshold. Oral beta-blocker should start within 24 h in stable patients. The 'truth wall' separates REAL contraindications (bradycardia/high-grade AV block, decompensated HF/cardiogenic shock, active bronchospasm, hypotension) from FAKE ones (stable COPD, PAD, diabetes). Mnemonic: 'beta-blockers: real reasons to hold are pump-and-pace problems, not COPD or diabetes.'

BOARD TRAP

WRONG answer: assuming every listed condition contraindicates beta-blockers. Discriminator — only hemodynamic/conduction/bronchospasm problems are real contraindications; stable COPD, PAD, and diabetes are pseudo-contraindications and should NOT cause you to withhold a mortality-reducing drug.

8. RED real contraindications

WHAT YOU SEE

'red = REAL CONTRAINDICATIONS' header

WHAT IT ENCODES

GENUINE contraindications to early beta-blockade in MI: bradycardia (HR <50–60) or PR prolongation/2nd–3rd-degree AV block, signs of acute heart failure or cardiogenic shock (Killip class III–IV), hypotension (SBP low), and active severe bronchospasm. In these settings early IV/oral beta-blockade can precipitate shock, asystole, or respiratory failure (COMMIT trial: early IV metoprolol increased cardiogenic shock in unstable patients). Mnemonic: 'hold the beta-blocker if the pump, the pacemaker, or the lungs are failing.'

BOARD TRAP

WRONG answer: treating red-listed items (heart block, shock, active asthma) as safe to give beta-blockers. Discriminator — COMMIT showed early beta-blockade harms hemodynamically unstable patients; these are true contraindications and you wait until the patient is stabilized.

9. Heart block / PR long / HR <50

WHAT YOU SEE

Red panel 'PR interval', 'AV block', '<50 BPM'

WHAT IT ENCODES

Bradycardia (HR <50–60) or high-degree AV block (PR prolongation, 2nd-degree Mobitz II, or 3rd-degree block) is a REAL contraindication — beta-blockers further suppress SA/AV nodal conduction and can cause symptomatic bradycardia or complete heart block/asystole. Inferior MIs (RCA, which supplies the AV node in ~90%) are especially prone to AV block. Mnemonic: 'slow heart or blocked node → no beta-blocker.'

BOARD TRAP

WRONG answer: giving a beta-blocker to a patient with complete (3rd-degree) heart block 'because beta-blockers help in MI.' Discriminator — nodal-blocking drugs are contraindicated in high-grade AV block; they can worsen the block to asystole. Stabilize conduction (atropine/pacing) first.

10. Cardiogenic shock / Killip III–IV

WHAT YOU SEE

Red panel 'CARDIOGENIC SHOCK / KILLIP III–IV'

WHAT IT ENCODES

Acute decompensated heart failure or cardiogenic shock (Killip class III = pulmonary edema, IV = shock) is a REAL contraindication to EARLY beta-blockade — the negative inotropy reduces an already-failing cardiac output and can precipitate or deepen shock. COMMIT showed early IV metoprolol increased cardiogenic-shock deaths in such patients. Beta-blockers are introduced LATER, once euvolemic and stable, where they then improve long-term HFrEF survival. Mnemonic: 'shock now = beta-blocker later.'

BOARD TRAP

WRONG answer: giving a beta-blocker to relieve acute cardiogenic shock/pulmonary edema. Discriminator — in acute decompensation beta-blockade reduces output and worsens shock; the survival benefit of beta-blockers in HFrEF applies only once the patient is compensated, not during acute shock.

11. Active asthma/bronchospasm

WHAT YOU SEE

Red panel 'ASTHMA / BRONCHOSPASM'

WHAT IT ENCODES

ACTIVE, severe bronchospasm (acute asthma attack) is a real contraindication — beta-blockade (especially non-selective) can trigger bronchoconstriction via β_2 blockade. However, STABLE asthma/COPD is only a relative concern, and a cardioselective (β_1) agent (metoprolol, bisoprolol, atenolol) is generally tolerated. Mnemonic: 'active wheeze = wait; stable lungs = cardioselective is fine.'

BOARD TRAP

WRONG answer: stating beta-blockers are absolutely safe during an active asthma attack. Discriminator — active bronchospasm IS a real contraindication, but the common board trap also runs the other way: STABLE COPD/asthma is NOT a contraindication and a cardioselective beta-blocker should be used rather than withheld.

12. GREEN fake contraindications

WHAT YOU SEE

'green = do NOT contraindicate' header

WHAT IT ENCODES

FAKE (pseudo-)contraindications that should NOT make you withhold a beta-blocker: STABLE COPD/asthma (use cardioselective β_1 agent), peripheral arterial disease (PAD), and DIABETES. Fear of masking hypoglycemia symptoms in diabetics is overstated and outweighed by the strong post-MI/HFrEF mortality benefit. Mnemonic: 'COPD (stable), PAD, Diabetes = green light for beta-blockers.'

BOARD TRAP

WRONG answer: withholding a beta-blocker post-MI because the patient has stable COPD or diabetes. Discriminator — these are pseudo-contraindications; denying a beta-blocker for them deprives the patient of a proven mortality benefit. Use a cardioselective agent and proceed.

13. Beta-blocker reduces mortality

WHAT YOU SEE

Green IV/oral beta-blocker, 'HR ↓ / BP ↓'

WHAT IT ENCODES

ORAL beta-blocker started within 24 h (in hemodynamically stable patients without contraindication) reduces post-MI mortality, reinfarction, and ventricular arrhythmia by cutting HR/BP/O₂ demand and raising the VF threshold. AVOID routine early IV beta-blockade in unstable patients (COMMIT harm). Typical agents: metoprolol tartrate/succinate, carvedilol, bisoprolol. Mnemonic: 'oral early, IV cautiously, always reassess hemodynamics.'

BOARD TRAP

WRONG answer: stating beta-blockers don't affect post-MI mortality, OR giving aggressive early IV beta-blockade to everyone. Discriminator — oral beta-blockade in stable patients clearly lowers mortality; but COMMIT showed routine early IV metoprolol increased cardiogenic shock — timing and patient stability matter.

14. Fake contraindications bin

WHAT YOU SEE

Trash bin 'FAKE CONTRAINDICATIONS'

WHAT IT ENCODES

Discard the pseudo-contraindications (stable COPD, PAD, diabetes, well-controlled HFrEF) — don't reflexively withhold a beta-blocker for a relative concern when the mortality benefit is large. The discipline: distinguish ABSOLUTE contraindications (treat as hard stops) from RELATIVE ones (proceed, often with a cardioselective agent). Mnemonic: 'trash the fake reasons; respect the real ones.'

BOARD TRAP

WRONG answer: treating every relative concern as an absolute contraindication and denying therapy. Discriminator — over-applying contraindications withholds a life-saving drug; only true hemodynamic/conduction/active-bronchospasm issues are absolute, and even those are often temporary (re-attempt once stable).

15. ACE inhibitor reduces remodeling

WHAT YOU SEE

Added card beside the beta-blocker wall: 'ACEi/ARB — START WITHIN 24H' — the drug that protects the infarcted ventricle's shape from remodeling.

WHAT IT ENCODES

NEW: An ACE inhibitor (or ARB if intolerant) started within 24 h reduces post-MI mortality and adverse LV remodeling, with the greatest benefit in ANTERIOR MI, reduced EF (<40%), heart failure, diabetes, or hypertension. It blunts the renin-angiotensin-driven dilation/scarring of the infarcted ventricle (SAVE, AIRE, ISIS-4). Watch for hypotension, hyperkalemia, and rising creatinine. Mnemonic: 'ACEi protects the pump's shape.'

BOARD TRAP

WRONG answer: omitting an ACE inhibitor after an anterior STEMI with reduced EF. Discriminator — ACEi is a class-I mortality-reducing adjunct that prevents remodeling; it is especially indicated (not optional) with anterior MI, EF<40%, HF, DM, or HTN.

16. High-intensity statin (all MI)

WHAT YOU SEE

Added card near the truth wall: 'HIGH-INTENSITY STATIN — EVERY MI' — the plaque-stabilizer every infarct patient gets regardless of baseline LDL.

WHAT IT ENCODES

NEW: EVERY MI patient gets a high-intensity statin (atorvastatin 80 mg or rosuvastatin 20–40 mg) started early, regardless of baseline LDL — for ≥50% LDL reduction plus plaque-stabilizing, anti-inflammatory (pleiotropic) effects (PROVE-IT, MIRACL). Secondary-prevention LDL target is <70 mg/dL (often <55). Mnemonic: 'every infarct earns an 80 of atorva.'

BOARD TRAP

WRONG answer: starting a moderate-intensity statin (or deferring it until lipids return) after MI. Discriminator — high-intensity statin is indicated for all ASCVD irrespective of LDL level, started in-hospital; under-dosing or waiting for lipid panels is a classic error.

17. Aldosterone antagonist if EF<40 + HF/DM

WHAT YOU SEE

Added lower-panel card: 'EPLERENONE — EF<40% + HF or DIABETES' — the aldosterone blocker reserved for the weak-pump-plus-HF/DM patient.

WHAT IT ENCODES

NEW: Add an aldosterone antagonist (eplerenone, or spironolactone) post-MI when EF <40% AND there is heart failure OR diabetes, already on an ACEi/ARB + beta-blocker — it reduces mortality (EPHESUS trial) by blocking aldosterone-driven fibrosis/remodeling and arrhythmia. Eplerenone is preferred (fewer antiandrogen side effects than spironolactone). Monitor potassium and renal function; avoid if K⁺ >5.0 or significant CKD. Mnemonic: 'low EF + (HF or DM) → eplerenone.'

BOARD TRAP

WRONG answer: giving an aldosterone antagonist to a post-MI patient with normal EF, or starting it without checking potassium/renal function. Discriminator — the indication is specifically EF<40% with HF or diabetes (EPHESUS); using it broadly or ignoring hyperkalemia risk is unsafe and not evidence-based.